

Machine Learning ('S403011')

Mock Exam

Multiple choice

Each multiple-choice question can have several correct answers.

1. For which of the following does normalizing your input features influence the predictions?
 - ☐ Multiple linear regression.
 - ☐ Support vector machines.
 - ☐ Lasso regularized regression.
 - ☐ Classification and regression trees.
2. What is usually true about increasing the k in k -Nearest-Neighbors?
 - ☐ The decision boundary tends to get smoother.
 - ☐ The bias tends to increase.
 - ☐ The variance tends to increase.
3. What strategies can help reduce overfitting in classification trees?
 - ☐ Enforce a minimum number of samples in each leaf.
 - ☐ Make sure each leaf node is one pure class.
 - ☐ Enforce a minimal depth for the tree.
 - ☐ Pruning.
4. Which of the followings statements are true about generative models?
 - ☐ They can be used for classification.
 - ☐ Linear discriminant analysis is a generative model.
 - ☐ They model the joint distribution of (X, Y) .
 - ☐ Logistic regression is a generative model.
5. Which of the following methods can achieve zero training error on any data set?
 - ☐ k -Nearest-Neighbors with $k = n$, where n denotes the size of the training sample.
 - ☐ Linear discriminant analysis (LDA).
 - ☐ Logistic regression.
 - ☐ Classification tree.

6. Gradient descent using minibatch optimization

- ☐ always finds the global minimum of the training loss function;
- ☐ always descends the training loss function in the optimal direction;
- ☐ descends the training loss function in small steps;
- ☐ is used for large training data sets.

7. If we had infinite data and infinitely fast computers, k -Nearest-Neighbors would be the only algorithm we would study in this class and use in practice.

- ☐ True.
- ☐ False.

8. Which of the following is an effective way of reducing overfitting in neural networks?

- ☐ Increasing the number of layers.
- ☐ Using convolutional layers.
- ☐ ℓ_2 -regularization.
- ☐ Dropout.

9. Neural networks:

- ☐ optimize a convex cost function;
- ☐ always output values between 0 and 1;
- ☐ can be used for regression as well as classification;
- ☐ can overfit the training data.

10. Which of the following methods can achieve zero training error on any linearly separable data set?

- ☐ 15-Nearest-Neighbors.
- ☐ Maximal margin classifier.
- ☐ Neural network.
- ☐ Classification tree.

11. In neural networks, nonlinear activation functions such as sigmoid and ReLU:

- ☐ are used to avoid overfitting;
- ☐ help to learn nonlinear decision boundaries;
- ☐ are used directly on the input data;
- ☐ always output values between 0 and 1.

12. The (linear) support vector classifier...

- ☐ gives a hyperplane as decision boundary.
- ☐ is fitted by maximum likelihood estimation.
- ☐ models the predictors by Gaussian distributions.
- ☐ models the posterior class probability.

13. Which of the following potentially reduces the bias of the predictive model?

- ☐ A larger penalization parameter in lasso regression.
- ☐ More trees in RandomForest.
- ☐ Smaller k in k -Nearest-Neighbors.
- ☐ Using more predictors in a linear model.

14. Logistic regression...

- ☐ can only be used with quantitative predictors.
- ☐ is fitted by maximum likelihood estimation.
- ☐ is a linear classification method.
- ☐ will never overfit the training data.

15. What is usually true if you grow a regression tree deeper and deeper?

- ☐ The training error decreases to zero.
- ☐ The test error first decreases and then increases.
- ☐ The training and test errors both decrease monotonically.
- ☐ The bias of the model decreases.

Problem 1

```
1 # Generate data
2 np.random.seed(42)
3 n = 1000
4 Id = np.array([[1/4, 0], [0, 1/4]])
5 x = np.concatenate(
6     [np.random.multivariate_normal(mean=[-1.5, 0], cov=Id, size=n // 2),
7      np.random.multivariate_normal(mean=[1.5, 0], cov=Id, size=n // 2)]
8 )
9 y = np.concatenate(
10    [np.repeat(0, repeats=n // 2),
11     np.repeat(1, repeats=n // 2)])
```

Figure 1: A python code chunk and its output.

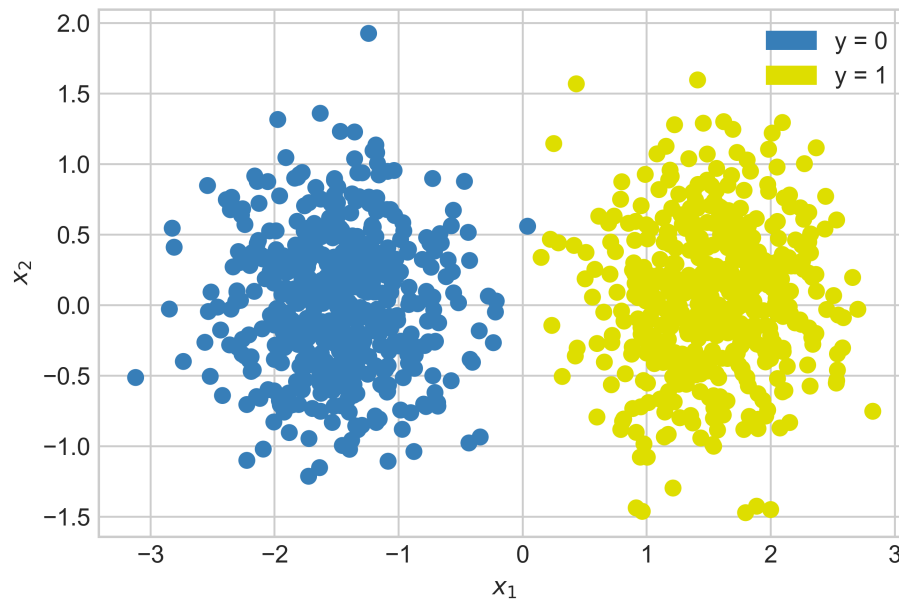


Figure 2: Training data from two different classes $y \in \{0, 1\}$ and two predictors x_1 and x_2 .

Figure 2 shows the training data $(\mathbf{x}_i, y_i)$, $\mathbf{x}_i \in \mathbb{R}^2, y_i \in \{0, 1\}, i = 1, \dots, n$, that has been generated according to the code in Figure 1. In this exercise we consider a binary classification problem.

- Write down the general formula for the classification method “logistic regression” and explain the quantities that appear. Write down how a new input $\mathbf{x} \in \mathbb{R}^2$ is classified.
- How many parameters has a logistic regression in the above example in Figure 2 and what method could be used to find the best parameters?
- Name two other classification methods that result in linear decision boundaries. Do all these linear methods give the same decision boundary when fitted to the above data? Explain why or why not.

- (d) Explain the k -Nearest-Neighbors classification method and write down how a new input $\mathbf{x} \in \mathbb{R}^2$ is classified. Intuitively, what is the role of k ?
- (e) Draw a schematic graph of the training and test errors for the k -Nearest-Neighbors method for the above data set with $k = 1, \dots, n$ on the x -axis and the errors on the y -axis. Explain your graph.

Problem 2

1. We fit SVM on the first two predictors of the Iris dataset. We use the radial basis function as a kernel. Each subplot (from A to D) corresponds to a very large or small value of γ and C . For each subplot, state whether the value of γ and C is large or small. Motivate your choice.

Hint: Recall that γ is the kernel coefficient of the radial basis function and C is the SVM regularization tuning parameter. The tuning parameter C here is inversely proportional to the “budget” seen during the lectures.

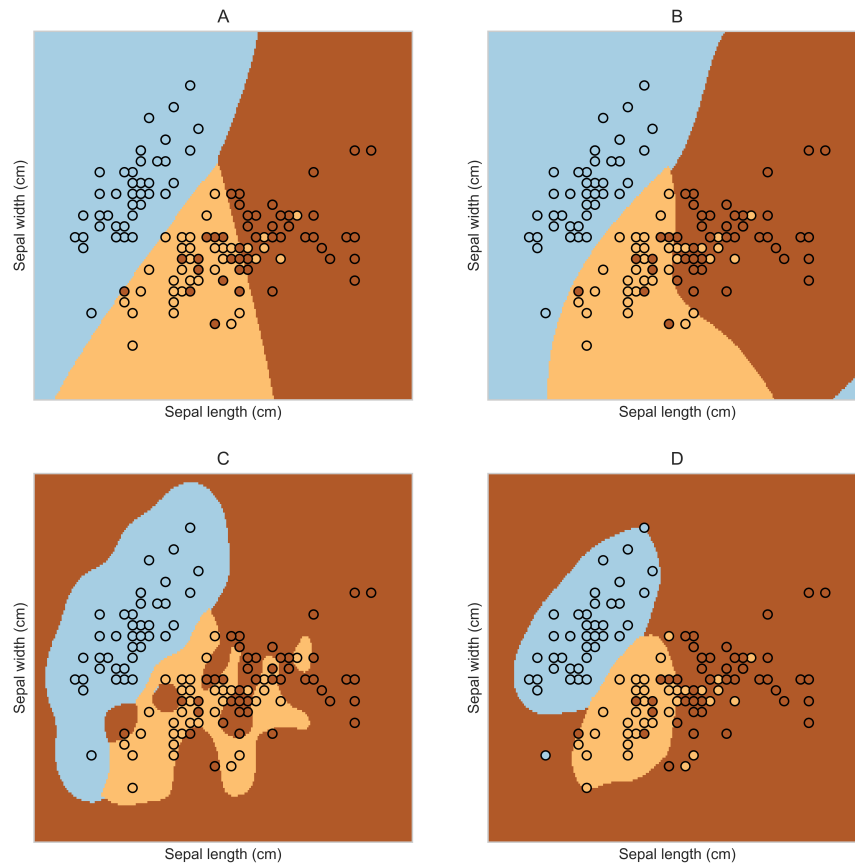


Figure 3: SVM fitted with different γ and C .



2. In the code of Figure 4, we perform a simple transform-fit-predict pipeline. However, there is a mistake that negatively affects the accuracy. Explain the problem.

```
1 # Transform-Fit-Predict pipeline
2 scaler = StandardScaler()
3 svm = SVC()
4
5 X_train_scaled = scaler.fit_transform(X_train)
6 svm.fit(X_train_scaled, y_train)
7
8 X_test_scaled = scaler.fit_transform(X_test)
9 y_test_pred = svm.predict(X_test_scaled)
```

Figure 4: Transform-fit-predict pipeline in python.



3. We fit boosting and random forest on the same dataset, for different number of iterations. At each iteration, we evaluate the performance of both methods on both the training and the test datasets. Figure 5 shows the four error curves obtained for boosting/random forest on the train/test dataset. Associate each error curve to the corresponding model (boosting or random forest) and dataset (train or test). Motivate your answer.

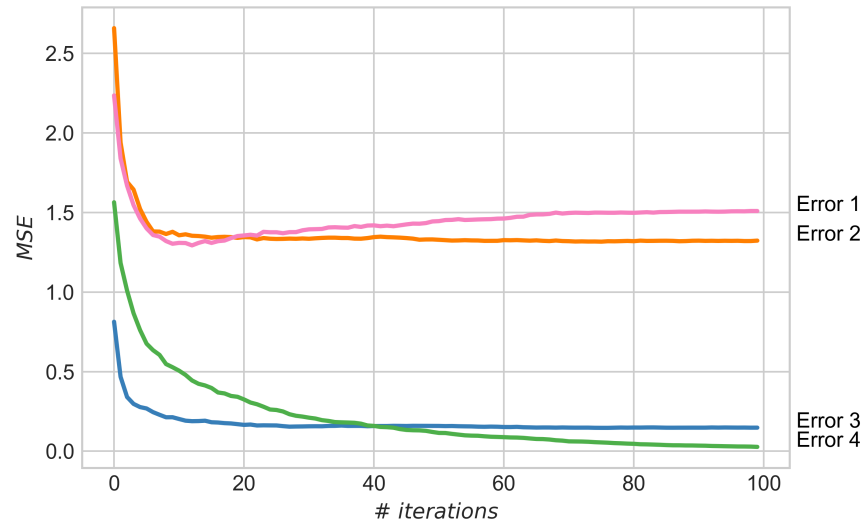


Figure 5: Error curves.



Problem 3

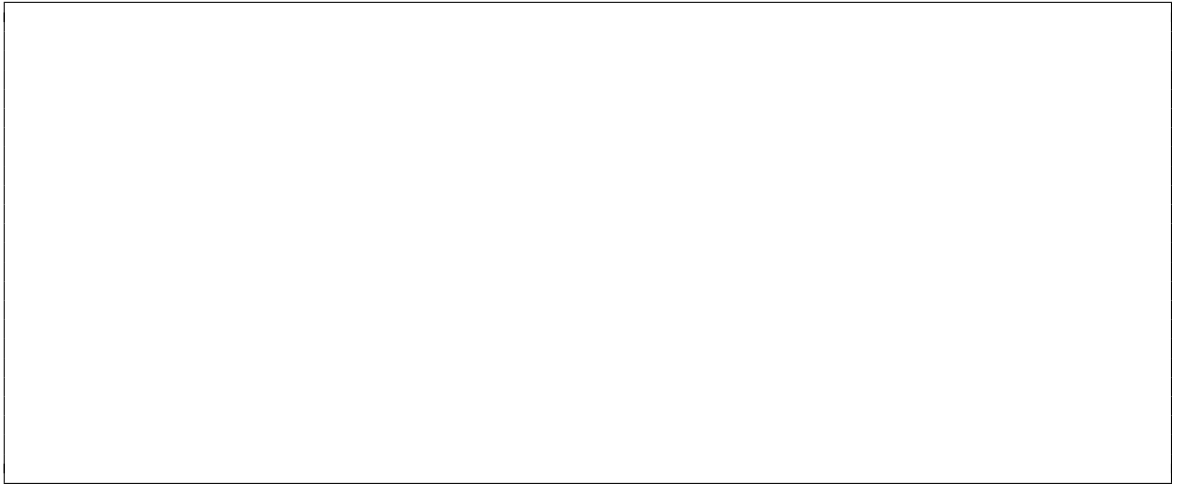
1. Suppose that we need to construct a model to recognise handwritten digits (0 to 9), which are given as 28 by 28 pixels grayscale images (which you can each see as a 28 by 28 matrix). Describe the dimensions of the classification task. In particular, discuss the number of features p and describe the response space (values that the response variable can take).

2. Figure 6 shows a possible model for the digits recognition task. Give the number of trainable parameters and the output shape of each layer (lines 8–13 of the code) from the neural network.

```
1 input_shape = (28, 28, 1)
2 X_tr_dl = np.expand_dims(X_train, -1)
3 X_te_dl = np.expand_dims(X_test, -1)
4 y_tr_long = keras.utils.to_categorical(y_train, n_class)
5 y_te_long = keras.utils.to_categorical(y_test, n_class)
6 modelnn = keras.Sequential([
7     keras.layers.InputLayer(input_shape=input_shape),
8     keras.layers.Conv2D(20, (5, 5), activation='relu'),
9     keras.layers.MaxPooling2D((2, 2)),
10    keras.layers.Flatten(),
11    keras.layers.Dense(100, activation='relu'),
12    keras.layers.Dropout(0.5),
13    keras.layers.Dense(n_class, activation="softmax")
14 ])
15 modelnn.compile(loss="categorical_crossentropy", optimizer="adam", metrics=["accuracy"])
16 modelnn.fit(X_tr_dl, y_tr_long, batch_size=128, epochs=15, verbose=0)
17 score = modelnn.evaluate(X_te_dl, y_te_long, verbose=0)
18 print("Validation loss:", score[0])
19 print("Validation accuracy:", score[1])
```

Validation loss: 0.03045995719730854
Validation accuracy: 0.9908999800682068

Figure 6: A python code chunk and its output.



3. What is the main advantage of convolutional neural networks compared to dense neural networks, **on image datasets**?

